

Derivation of every equation in `sw_sn_driver.py`

Full shallow water on a finite-amplitude meander, channel-fitted (s, n)

numerical/dedalus_meander_full_SW

This note derives, in order, every formula that appears in `sw_sn_driver.py`. Each section names the code symbol it produces.

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1 Why (s, n) , and the metric $\sigma = 1 + nC$

The channel meanders with *finite* amplitude, so its banks are not straight lines and a Chebyshev grid cannot be laid on them directly. We therefore use **channel-fitted** coordinates: s = arc length along the centreline, n = signed distance across it. In (s, n) the channel is a straight strip $n \in [-b, b]$ whatever the planform does.

Let the centreline be $\mathbf{r}_c(s)$ with unit tangent $\hat{\mathbf{t}}(s)$ and unit normal $\hat{\mathbf{n}}(s)$, and let $C(s)$ be its curvature. A point of the domain is

$$\mathbf{r}(s, n) = \mathbf{r}_c(s) + n \hat{\mathbf{n}}(s). \quad (1)$$

With the Frenet relation $d\hat{\mathbf{n}}/ds = +C \hat{\mathbf{t}}$ (this *defines* our sign convention: n increases *away* from the centre of curvature; equivalently the direction angle is $\theta(s) = -\int^s C ds'$),

$$\frac{\partial \mathbf{r}}{\partial s} = \hat{\mathbf{t}} + n \frac{d\hat{\mathbf{n}}}{ds} = (1 + nC) \hat{\mathbf{t}}, \quad \frac{\partial \mathbf{r}}{\partial n} = \hat{\mathbf{n}}, \quad (2)$$

so the coordinates are orthogonal with scale factors

$$\boxed{h_s = \sigma \equiv 1 + nC(s), \quad h_n = 1.} \quad (\rightarrow \text{sigma_metric, sig})$$

σ is the local stretching of an along-channel line at offset n : the outer bank (n with the sign of C) is longer, the inner bank shorter.

Validity. $\sigma > 0$ is required or the mapping folds. Since $|n| \leq b$, this is $|C|b < 1$: the bend radius must exceed the half-width. The test `tests/test_base_profiles.py` asserts $\min \sigma > 0$.

Sign convention (important). Mirroring the channel ($C \rightarrow -C$) maps the problem to itself with $n \rightarrow -n$, so the linear growth rate $\sigma(k)$ and phase speed $c(k)$ are *unchanged*; only the labelling of “outer” vs “inner” bank flips. The convention above is the one used by the solver ($\sigma = 1 + nC$); the lab-frame renderer must integrate $\theta = -\int C ds$ to match it.

2 Shallow-water equations in (s, n)

2.1 The starting (Cartesian) system

Non-rotating depth-averaged shallow water over a bed of still-water depth H , free surface η , total depth $h = \eta + H$:

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -g \nabla \eta + \mathbf{F}, \quad \partial_t \eta + \nabla \cdot (h \mathbf{u}) = 0. \quad (3)$$

The pressure is hydrostatic, $p = \rho g(\eta - z)$, so the horizontal pressure gradient is exactly $-g \nabla \eta$; this is the $L \gg H$ (long-wave) limit. Keeping $\partial_t \eta$ is what admits **gravity waves**; the rigid-lid packages in ../ drop it and are left with the vortical branch only.

2.2 Curvilinear operators

For orthogonal coordinates with scale factors $(h_s, h_n) = (\sigma, 1)$:

$$\nabla \phi = \frac{1}{\sigma} \partial_s \phi \hat{\mathbf{t}} + \partial_n \phi \hat{\mathbf{n}}, \quad (4)$$

$$\nabla \cdot \mathbf{F} = \frac{1}{\sigma} \left[\partial_s F_s + \partial_n (\sigma F_n) \right], \quad (5)$$

$$\zeta = (\nabla \times \mathbf{u}) \cdot \hat{\mathbf{z}} = \frac{1}{\sigma} \left[\partial_s u_n - \partial_n (\sigma u_s) \right]. \quad (6)$$

The material acceleration acquires *metric* (centrifugal/Coriolis-like) terms from the rotation of the basis vectors. With $D/Dt = \partial_t + (u_s/\sigma)\partial_s + u_n\partial_n$ and $\partial h_s/\partial n = C$, $\partial h_n/\partial s = 0$, the standard result is

$$a_s = \frac{Du_s}{Dt} + \frac{C u_s u_n}{\sigma}, \quad a_n = \frac{Du_n}{Dt} - \frac{C u_s^2}{\sigma}. \quad (7)$$

2.3 The system solved

Substituting (5)–(7) into (3):

$$\partial_t \eta + \frac{1}{\sigma} [\partial_s(hu_s) + \partial_n(\sigma h u_n)] = 0, \quad (8)$$

$$\partial_t u_s + \frac{u_s}{\sigma} \partial_s u_s + u_n \partial_n u_s + \underbrace{\frac{C u_s u_n}{\sigma}}_{\text{bend coupling}} = -\frac{g}{\sigma} \partial_s \eta - r_s u_s + \nu \nabla^2 u_s, \quad (9)$$

$$\partial_t u_n + \frac{u_s}{\sigma} \partial_s u_n + u_n \partial_n u_n - \underbrace{\frac{C u_s^2}{\sigma}}_{\text{centrifugal}} = -g \partial_n \eta - r_n u_n + \nu \nabla^2 u_n. \quad (10)$$

The term $-C u_s^2/\sigma$ is the whole reason for going to (s, n) : it is the centrifugal acceleration of water rounding a bend, and it is what tilts the free surface up against the outer bank (§4). No straight-channel model contains it.

3 Non-dimensionalisation: where the Froude number enters

Scale lengths by the half-width b , speeds by the centreline speed $U_c = U_0 + \Delta$, time by b/U_c , and depths/ η by the mean depth H_0 . The only place g survives is the pressure gradient, where it appears as

$$\frac{gH_0}{U_c^2} = \frac{1}{F^2}, \quad \boxed{F \equiv \frac{U_c}{\sqrt{gH_0}}, \quad g_{\text{eff}} = \frac{1}{F^2}} \quad (\rightarrow \text{g_eff})$$

so the non-dimensional gravity-wave speed is $\sqrt{g_{\text{eff}}H_0} = 1/F$.

Why F must be specified, and why one value proves nothing. F is an *input* (a property of the river), exactly like a Reynolds number: without it the system (8)–(10) is not defined. At any F *both* families exist — gravity waves at speed $1/F$ and the vortical branch at $O(U)$. Which one the bank couples to is decided by *measurement*: compare c_{meander} with $1/F$, and above all *vary* F . If the meander rode the gravity branch, tripling $1/F$ would move σ, c . It does not (`sweep_dispersion.py` $\rightarrow$ `fig01`), which is the evidence that the meander is the vortical/Rossby wave.

4 The base state

4.1 Parabolic jet and the constant channel- β

$$\boxed{\bar{U}_s(n) = U_0 + \Delta \left(1 - \frac{n^2}{b^2}\right)} \quad (\rightarrow \text{ubar_s})$$

so $\bar{U}_s(0) = U_0 + \Delta$ (centre) and $\bar{U}_s(\pm b) = U_0$ (bank edge), and

$$\partial_n \bar{U}_s = -\frac{2\Delta n}{b^2} \quad (\rightarrow \text{ubar_s_n}), \quad \partial_n^2 \bar{U}_s = -\frac{2\Delta}{b^2} = \text{const} \quad (\rightarrow \text{ubar_s_nn}). \quad (11)$$

This constant is the load-bearing property. From (6) with $u_n = 0$ and (for the moment) $C = 0$, the base vorticity is $\bar{\zeta} = -\partial_n \bar{U}_s$, hence

$$\boxed{\frac{\partial \bar{\zeta}}{\partial n} = -\partial_n^2 \bar{U}_s = \frac{2\Delta}{b^2} = \text{constant}} \quad (12)$$

— a uniform cross-channel gradient of background vorticity. This is the exact analogue of the planetary β : it is the restoring mechanism that makes a *vortical (Rossby-type) wave* possible in a river. A parabolic jet is thus not cosmetic; it is what gives the channel a constant β . (For $C \neq 0$ the same calculation gives $\bar{\zeta} = -\partial_n \bar{U}_s - C \bar{U}_s / \sigma$, a curvature correction.)

4.2 Base curvature = the finite background meander

$$\boxed{\bar{C}(s) = \bar{C}_{\text{amp}} \cos(k_m s), \quad \bar{C}_{\text{amp}} = A_{\text{bank}} k_m^2 \text{ (default)}} \quad (\rightarrow \text{cbar})$$

For a centreline of lateral amplitude A_{bank} and wavenumber k_m , the small-slope curvature is $-\partial_s^2 (A_{\text{bank}} \cos k_m s) = A_{\text{bank}} k_m^2 \cos k_m s$; `Cbar_amp` overrides this to set a genuinely tight bend directly. **A_{bank} is the fixed background channel shape — it is *not* the perturbation seed A_0 of §10, and it does not evolve.**

4.3 Superelevation

For a steady base with $\bar{U}_n = 0$ and $\partial_s(\cdot) = 0$, the n -momentum (10) reduces to a balance between the centrifugal term and the pressure gradient:

$$\boxed{g \partial_n \bar{\eta} = \frac{\bar{C} \bar{U}_s^2}{\bar{\sigma}}} \quad (\rightarrow \text{etabar})$$

integrated as $\bar{\eta}(s, n) = \frac{\bar{C}(s)}{g} \int_0^n \frac{\bar{U}_s^2(n')}{1 + n' \bar{C}(s)} dn'$ (zero on the centreline). The water surface tilts *up towards the outer bank* — the superelevation. Note $\bar{C} \rightarrow 0 \Rightarrow \bar{\eta} \rightarrow 0$ (asserted in the test).

4.4 Discharge conservation

The base continuity (8) with $\bar{U}_n = 0$ requires $\partial_s(\bar{h} \bar{U}_s) = 0$: the transport is the same through every cross-section. With a cross-channel-only bed this is automatic ($\bar{U}_s = \bar{U}_s(n)$).

What sustains the base. A parallel jet with bottom drag is not a steady solution unless something drives it (the downhill slope). As in every frozen-jet stability analysis we *prescribe* $\bar{U}_s$ and absorb the residual into an implicit steady forcing; it cancels from the perturbation equations below.

5 Linearisation

Write $u_s = \bar{U}_s + u'_s$, $u_n = u'_n$, $\eta = \bar{\eta} + \eta'$, $h = \bar{h} + \eta'$, $C = \bar{C} + C'$, and drop products of primes.

5.1 Continuity

$\partial_s(h u_s) = \partial_s(\bar{h} \bar{U}_s) + \bar{h} \partial_s u'_s + \bar{U}_s \partial_s \eta'$, where the first term vanishes (discharge), and $\partial_n(\sigma h u_n) \rightarrow \partial_n(\bar{\sigma} \bar{h} u'_n)$. Hence

$$\bar{\sigma} \partial_t \eta' + \bar{h} \partial_s u'_s + \bar{U}_s \partial_s \eta' + \partial_n(\bar{\sigma} \bar{h} u'_n) = 0, \quad (\rightarrow \text{eq. 1 in build_ivp_SW})$$

already multiplied by $\bar{\sigma}$ (§6).

5.2 s -momentum

$\frac{u_s}{\sigma} \partial_s u_s \rightarrow \frac{\bar{U}_s}{\bar{\sigma}} \partial_s u'_s$; $u_n \partial_n u_s \rightarrow u'_n \partial_n \bar{U}_s$; $\frac{C u_s u_n}{\sigma} \rightarrow \frac{\bar{C} \bar{U}_s u'_n}{\bar{\sigma}}$. Multiplying by $\bar{\sigma}$ collects the two u'_n terms into one coefficient:

$$\bar{\sigma} \partial_t u'_s + \bar{U}_s \partial_s u'_s + \underbrace{(\bar{\sigma} \partial_n \bar{U}_s + \bar{C} \bar{U}_s)}_{\rightarrow \text{coefUn}} u'_n + g \partial_s \eta' + \bar{\sigma} r_s u'_s - \bar{\sigma} \nu \nabla^2 u'_s = 0. \quad (\rightarrow \text{eq. 2})$$

The u'_n coefficient is the cross-channel advection of background momentum — proportional to the base vorticity — and is the term that carries the Rossby-type restoring.

5.3 n -momentum and the curvature feedback

Expanding $-C u_s^2 / \sigma$ with $C = \bar{C} + C'$, $\sigma = \bar{\sigma} + n C'$:

$$-\frac{C u_s^2}{\sigma} \simeq \underbrace{-\frac{\bar{C} \bar{U}_s^2}{\bar{\sigma}}}_{\text{base}} - \underbrace{\frac{2 \bar{C} \bar{U}_s u'_s}{\bar{\sigma}}}_{\rightarrow \text{twoCbUb}} - \underbrace{\frac{C' \bar{U}_s^2}{\bar{\sigma}}}_{\text{new bend}} + \frac{n C' \bar{C} \bar{U}_s^2}{\bar{\sigma}^2}. \quad (13)$$

The perturbed pressure term contributes $-\bar{\sigma} g \partial_n \eta' - n C' g \partial_n \bar{\eta}$, and using the base balance ($\rightarrow \text{etabar}$), $g \partial_n \bar{\eta} = \bar{C} \bar{U}_s^2 / \bar{\sigma}$, the last piece is $-n C' \bar{C} \bar{U}_s^2 / \bar{\sigma}$. Adding it to the $+n C' \bar{C} \bar{U}_s^2 / \bar{\sigma}^2$ above and using $\bar{\sigma} - n \bar{C} = 1$ leaves the remarkably clean forcing

$$\boxed{n\text{-momentum forcing} = \frac{C' \bar{U}_s^2}{\bar{\sigma}} \simeq C' \bar{U}_s^2.} \quad (14)$$

So the final equation, with the forcing moved to the left and $C' = -\partial_s^2 \zeta_c$ (§9):

$$\bar{\sigma} \partial_t u'_n + \bar{U}_s \partial_s u'_n - 2 \bar{C} \bar{U}_s u'_s + \bar{\sigma} g \partial_n \eta' + \bar{\sigma} r_n u'_n - \bar{\sigma} \nu \nabla^2 u'_n + \underbrace{\bar{U}_s^2 \partial_s^2 \zeta_c}_{\text{bend} \rightarrow \text{flow}} = 0. \quad (\rightarrow \text{eq. 3})$$

This is the coupling that closes the meander instability: a bend creates a centrifugal forcing, which drives the flow, which erodes the bank, which deepens the bend.

6 Why every equation is multiplied by σ

The raw equations carry $1/\sigma = 1/(1+nC)$, a *rational* function of n . Dedalus builds such a coefficient as a dense operator (its documentation warns about exactly this), destroying the banded structure. Multiplying each equation through by σ removes every $1/\sigma$ and leaves coefficients that are polynomial in n (products of σ , $\bar{h}$, $\bar{U}_s$), so the matrices stay banded. Multiplying an equation by a nonzero function is an exact, solution-preserving row operation. The one residual $1/\bar{\sigma}$ (in the C' forcing) is dropped at $O(n\bar{C}C')$, i.e. second order in the perturbation amplitude *and* the base curvature.

7 Bottom friction: the factor of two

Quadratic (Chézy) drag is $\mathbf{F} = -\frac{C_f}{h} |\mathbf{u}| \mathbf{u}$. With $\mathbf{u} = (\bar{U}_s + u'_s, u'_n)$ and $|\mathbf{u}| \simeq \bar{U}_s + u'_s$ (the u'_n contribution is second order):

$$|\mathbf{u}| u_s \simeq \bar{U}_s^2 + 2 \bar{U}_s u'_s \quad \Rightarrow \quad r_s = \frac{2 C_f \bar{U}_s}{\bar{h}} \quad (\rightarrow \text{rs_a}), \quad (15)$$

$$|\mathbf{u}| u_n \simeq \bar{U}_s u'_n \quad \Rightarrow \quad r_n = \frac{C_f \bar{U}_s}{\bar{h}} \quad (\rightarrow \text{rn_a}). \quad (16)$$

The drag is therefore **anisotropic**: streamwise perturbations are damped twice as fast as cross-stream ones, because they perturb $|\mathbf{u}|$ as well as the direction. (The same factor 2 appears in the “momentum” closure of the rigid-lid packages.) The base part $C_f \bar{U}_s^2 / \bar{h}$ belongs to the implicit driving balance and cancels.

8 Boundary conditions, and why there are four tau terms

The banks sit at fixed $n = \pm b$ in channel coordinates. Water cannot cross a solid bank, so

$$\boxed{u'_n(\pm b) = 0} \quad (\text{no penetration}). \quad (17)$$

This assumes *nothing* about the interior divergence: $\nabla \cdot \mathbf{u} \neq 0$ everywhere inside (that is what the free surface is for). In particular the bank is *not* treated as a material streamline — that would be a rigid-lid (non-divergent) assumption and is inconsistent with (8).

Lateral eddy viscosity $\nu \nabla^2$ makes each momentum equation second order in n , which (i) is physically reasonable for a river, (ii) makes the parabolic jet an actual steady solution, and (iii) gives a standard tau closure. A second-order operator needs one boundary condition per wall per component, so with free slip

$$\partial_n u'_s(\pm b) = 0 \quad (\text{free slip}), \quad (18)$$

the tau budget is 2 per viscous velocity component:

$$\underbrace{2}_{u'_s} + \underbrace{2}_{u'_n} + \underbrace{0}_{\eta'} = 4 \quad (\rightarrow \text{t_us1}, \text{t_us2}, \text{t_un1}, \text{t_un2}),$$

η' needing none because its equation contains no second n -derivative. Counting unknowns and equations: $\{u'_s, u'_n, \eta', \zeta_c\} + 4 \text{ tau} = 8$ and $3 \text{ PDE} + 2 \text{ no-penetration} + 2 \text{ free-slip} + 1 \text{ erosion} = 8$. The tau terms are lifted onto the last two Chebyshev modes, `lift(t, -1) + lift(t, -2)` on `ybasis.derivative_basis(2)`.

9 Erodible bank: the meander equation

The bank is a sediment boundary, not a material line: it recedes because the flow erodes it. Ikeda’s closure makes the migration rate proportional to the *near-bank excess longitudinal velocity*. For a **bend** the outer bank is fast and the inner bank slow, so it is the **antisymmetric** part that moves the centreline:

$$\boxed{\partial_t \zeta_c = E \cdot \frac{1}{2} [u'_s(+b) - u'_s(-b)]}, \quad E = \varepsilon U_0 \quad (\rightarrow \text{eq. 8, bank_E})$$

where $\zeta_c(s, t)$ is the lateral offset of the centreline (constant width).

Why antisymmetric. Using two independent banks with $\partial_t \xi_{\pm} = E u'_s(\pm b)$ instead makes the centreline $\frac{1}{2}(\xi_+ + \xi_-)$ respond to the *symmetric* part of the near-bank velocity. That is the channel-*widening* mode, not bending: in practice it gives $\sigma < 0$ (no meander instability). Parity matters.

Closing the loop. The migrating centreline changes the curvature,

$$C'(s, t) = -\partial_s^2 \zeta_c, \quad (19)$$

which re-enters the n -momentum as the forcing derived in §5.3. The cycle

$$\zeta_c \rightarrow C' \rightarrow \text{flow} \rightarrow u'_s(\pm b) \rightarrow \partial_t \zeta_c$$

is the meander instability. Its growth rate is what the IVP measures.

10 Seeding, and why the seed must be a domain mode

The base state is an exact steady solution: started there, nothing ever happens. Stability analysis therefore *requires* a perturbation, and its growth rate is the answer. We seed a pure bend with zero initial flow:

$$\zeta_c(s, 0) = A_0 \cos(k_{\text{eff}} s), \quad u'_s = u'_n = \eta' = 0. \quad (20)$$

Mode snapping. $\cos(ks)$ is periodic on $[0, L_s]$ only if $kL_s/2\pi$ is an integer. Otherwise the seed has a jump at the domain edge, whose spectral content spreads over many modes and grows as a *localized packet* instead of a clean single-wavelength meander. We therefore snap to the nearest exact mode:

$$m = \text{round}\left(\frac{kL_s}{2\pi}\right), \quad k_{\text{eff}} = \frac{2\pi m}{L_s}. \quad (21)$$

A_0 is a free constant. The system is linear, so if $(u'_s, u'_n, \eta', \zeta_c)$ solves it then so does $\lambda \times$ it. Only σ , c and the *ratios* between fields are physical; the overall size is set by whatever initial disturbance one assumes. This is why the movie may fix the overall constant once (final meander = $0.5b$) and apply it to every frame and every field — that is a re-choice of A_0 , not a distortion.

11 Measuring σ and c

The centreline is demodulated onto its seeded mode, $a(t) = \hat{\zeta}_c(m, t)$, and

$$\sigma = \frac{d}{dt} \ln |a|, \quad c = -\frac{1}{k} \frac{d}{dt} \arg a, \quad (\rightarrow \text{measure_sigma_c})$$

both fitted over the last two thirds of the run (dropping the initial adjustment, during which the flow spins up from rest). $\sigma > 0$ is an unstable meander; $c > 0$ means the bend train migrates downstream.

12 What the model deliberately does *not* contain

Strictly linear perturbation dynamics (no $\mathbf{u}' \cdot \nabla \mathbf{u}'$); no sediment (Exner) feedback — the bed H is prescribed, only the banks move; no rotation ($f = 0$, a river); no vertical structure, hence no helical secondary flow (it is depth-averaged away); and the base state is frozen (its steady residual is an implicit forcing). The lateral viscosity uses the flat Laplacian rather than the exact curvilinear vector Laplacian — it is a wall-closure and short-wave regulariser, small, and not part of the physics being measured.